

MOHAMED MAHMOUD SALEM

AI Engineering Student | Machine Learning | Deep Learning | NLP | Computer Vision | Generative AI

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PROFESSIONAL SUMMARY

Third-year Artificial Intelligence Engineering student with hands-on experience in Machine Learning, Deep Learning, NLP, Computer Vision, and Generative AI. Proficient in Python with practical experience building, evaluating, debugging, and optimizing AI models and applications. Experienced with Scikit-learn, TensorFlow, Keras, Hugging Face Transformers, LangChain, LLMs, RAG, and REST APIs through applied projects and technical training. Seeking an AI/ML Engineering internship to apply analytical problem-solving skills in a practical engineering environment.

EDUCATION

B.Eng. – Electrical and Electronics Engineering (Artificial Intelligence)

New Ismailia National University | Nov 2023 – Jul 2028 | Ismailia, Egypt | GPA: 3.15 / 4.0

WORK LIFE EXPERIENCE

Computer Vision Trainee | National Telecommunication Institute (NTI)

Jul 12 – Aug 6, 2026 | 120 Hours | Score: 96.5%

- Completed 120 hours of Computer Vision summer training, including 90 technical hours and 30 freelancing hours.
- Applied Computer Vision concepts and practical techniques through structured technical training.

Machine Learning Trainee | National Telecommunication Institute (NTI)

Jun 2025 | Ismailia, Egypt

- Completed 120 hours of applied Machine Learning training, developing and validating predictive models for customer churn, flight pricing, and salary estimation using Python and Scikit-learn.
- Applied preprocessing pipelines and evaluated models using accuracy, precision, recall, and F1-score.
- Collaborated on applied ML case studies and delivered technical presentations communicating model findings and results.

Data Analysis Trainee | Ministry of Communications and Information Technology (MCIT)

Aug 2025 | Ismailia, Egypt

- Cleaned, preprocessed, and analyzed real-world datasets using Python, Pandas, and NumPy.
- Built interactive Power BI and Excel dashboards to communicate data-driven business insights.

Generative AI Trainee | NVIDIA Deep Learning Institute (DLI) & ITI

Aug 2025 | Ismailia, Egypt

- Completed applied training covering Large Language Models (LLMs), prompt engineering, and Retrieval-Augmented Generation (RAG).
- Built LLM-based applications using REST API integration and evaluated prompt outputs for accuracy and relevance.

Deep Learning Trainee | Syndicate of Sub-Engineers in Cairo

Sep 2025 | Cairo, Egypt

- Built and evaluated neural networks and Convolutional Neural Networks (CNNs) for image classification.
- Applied structured preprocessing, model training, and performance evaluation methodologies.

PROJECTS

Multilingual Fake News Detection | NLP - Hugging Face - Transformers

- Fine-tuned pre-trained transformer models for fake-news classification in Arabic and English.
- Analyzed misclassified samples to identify failure patterns and improve model robustness.

Facial Emotion Recognition (FER) | TensorFlow - Keras - Transfer Learning

- Built a 7-class facial emotion classifier trained on more than 50,000 images using CNNs and transfer learning.
- Identified and resolved a training-configuration bug that was masking true validation performance.

Structural Crack Detection | CNN - TensorFlow - Keras - Computer Vision

- Developed a binary CNN classifier for structural crack detection using dataset preprocessing and augmentation.
- Evaluated model performance using precision, recall, and F1-score.

Customer Label Evaluation – DeepX Hackathon | NLP - Multilingual - Franco-Arabic

- Built an NLP pipeline for multilingual customer-feedback label evaluation, including Franco-Arabic text.
- Implemented intent classification and entity extraction for code-switched text.

Bemo – Conversational Chatbot | LangChain - LLMs - REST APIs - Python

- Developed a context-aware chatbot using LangChain and LLM APIs.

- Designed and validated conversational flows, intent handling, and dialogue-management logic.

Customer Churn Prediction | Python · Scikit-learn · ML Optimization

- Built a multi-algorithm classification pipeline for customer-churn prediction.
- Applied cross-validation and standard evaluation metrics to compare models and select the most reliable approach.

Bank Management System | Python · SQL · Database Design

- Built a full-stack banking system with integrated database design for account management and transaction tracking.

AI & Solar Energy Optimization

- Researching AI/ML techniques for solar-energy performance prediction and efficiency optimization.

CERTIFICATIONS & ACHIEVEMENTS

- Computer Vision | National Telecommunication Institute (NTI) | Jul–Aug 2026 | Score: 96.5%
- Machine Learning | National Telecommunication Institute (NTI) | Jun 2025
- Generative AI & LLMs | NVIDIA Deep Learning Institute (DLI) & ITI | Aug 2025
- Data Analysis | Ministry of Communications and Information Technology (MCIT) | Aug 2025
- Deep Learning | Syndicate of Sub-Engineers in Cairo | Sep 2025
- Hult Prize 2025 Egypt National Competition – Startup Team Member | 2025

VOLUNTEERING

Technical Head | Microsoft Student Club – SCU

Jan 2026 – Present

- Lead the technical team in organizing AI/ML workshops and technology sessions.
- Mentor members and produce technical content for student-focused initiatives.

Golden AI Member | IEEE Suez Canal University Student Branch

Feb 2026 – Jun 2026

- Recognized as a Golden AI Member in April 2026 for outstanding contribution.
- Delivered NLP and Computer Vision projects under expert mentorship while exploring Agentic AI workflows.

Golden HR | EYE – Ministry of Youth and Sports of Egypt

2025 – Present

- Managed HR and organizational operations for a national youth initiative.
- Developed formal documentation, including a monthly member-evaluation system for team-performance tracking.

TECHNICAL SKILLS

Programming: Python, C++

Machine Learning & AI: Machine Learning, Deep Learning, NLP, Computer Vision, Generative AI, LLMs, RAG, Agentic AI

Frameworks & Libraries: TensorFlow, Keras, Scikit-learn, Hugging Face Transformers, LangChain

Data & Tools: Pandas, NumPy, Power BI, Excel, Git, GitHub

APIs & Development: REST APIs, Pre-trained Models

SOFT SKILLS

Problem-Solving · Communication · Team Collaboration · Leadership · Adaptability

LANGUAGES

Arabic: Native / Bilingual | English: Professional Working Proficiency